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Glossary

See the Glossary for a list of commonly used terms and definitions.

Overview

Getting Started

Downloading and Installing

OS/Platform Differences

- Keyboard shortcuts

UI Overview

Scaling

“the whole UI scales with the window width, against a 3840 pixel reference. A window 3840 wide renders at scale 1.0, and a window 1280 wide renders everything, fonts included, at one third of that. Nothing is tied to the display DPI. If you work in a small window, raise UI Scale and Font Scale in Settings to compensate; they multiply on top of the automatic factor.”

Your First Project

Understanding File Types

Supporting Models

Supporting Overview

Basic Workflow

Features

Support Basics

- Horizontal line of layers

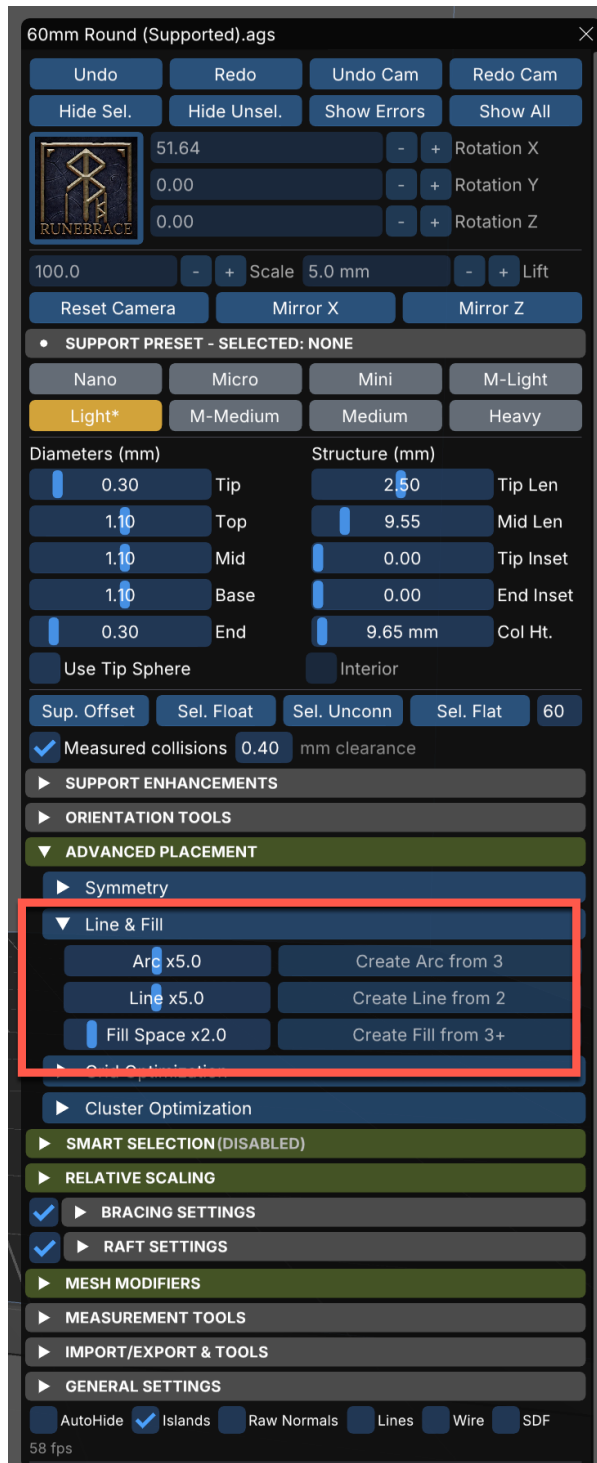
Manual Supports

Symmetry

Path Supports

Path supports

Settings controlling spacing of supports for the following features are located in the right menu under Advanced Placement => Line and Fill.

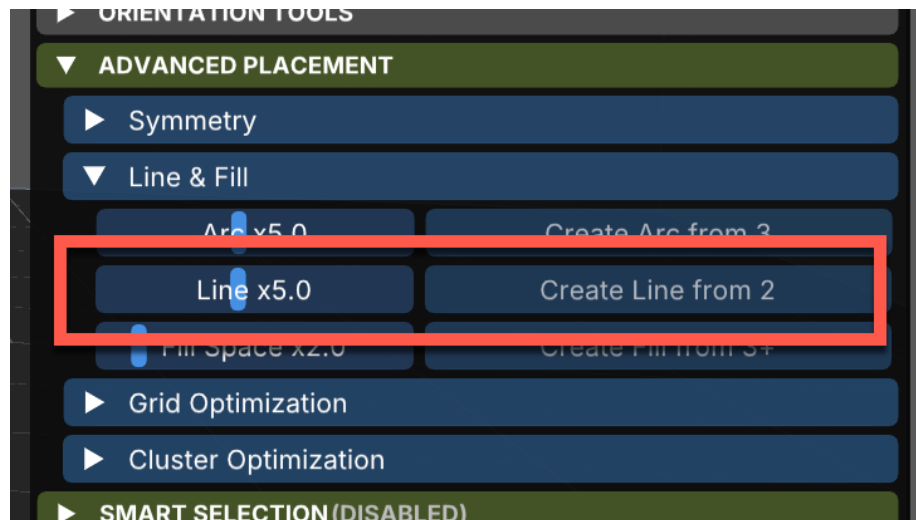


Line

“What happens when you draw support lines or arcs is that it simulates a mouse clicking on various points along this line (depending on the set distance) and performs a mouse raycast on the model. If the raycast hits the geometry and the face is not parallel to the build plate or tilted upwards, a support is placed. It is likely that there are some raycast misses, so the support cannot be placed and is skipped. It’s not a matter of whether the surface is flat... the geometry must be under the ray cast by the mouse and be a valid point for a support.”

Runebrace does its best to place supports equally along the line per the above paragraph.

Spacing for supports along the line is controlled by the right menu under Advanced Placement => Line & Fill => Line.



Note that there will be a gap or compression of supports along a line that isn't evenly divisible by the spacing setting.

For example, two supports 13 mm apart are selected, and Line spacing is 5.0. 13 isn't evenly divisible by 5, so the algorithm will start at the first support and place supports at five and ten mm away. There will be a gap of 3mm from the

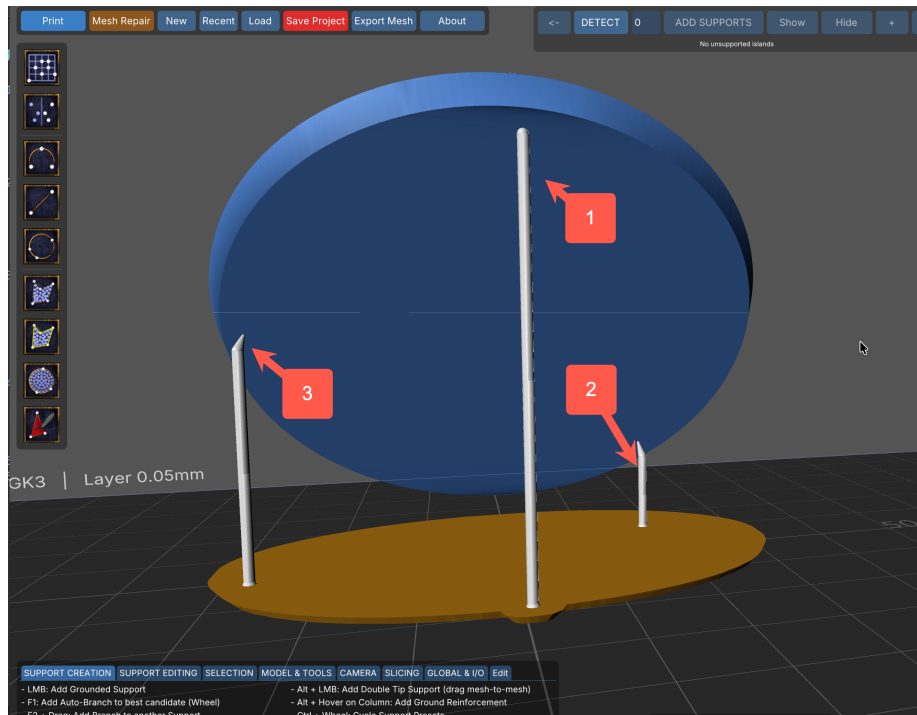
Arc

Circle

Fill Basics

We'll start with a basic example, then show a more complex, practical example after. Triangle Fills are covered separately below.

All examples in this section start from this figure, a round base with three supports selected in the order they're labeled.

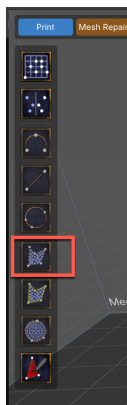


Each feature below will show results starting from the above figure.

Polygon Fill

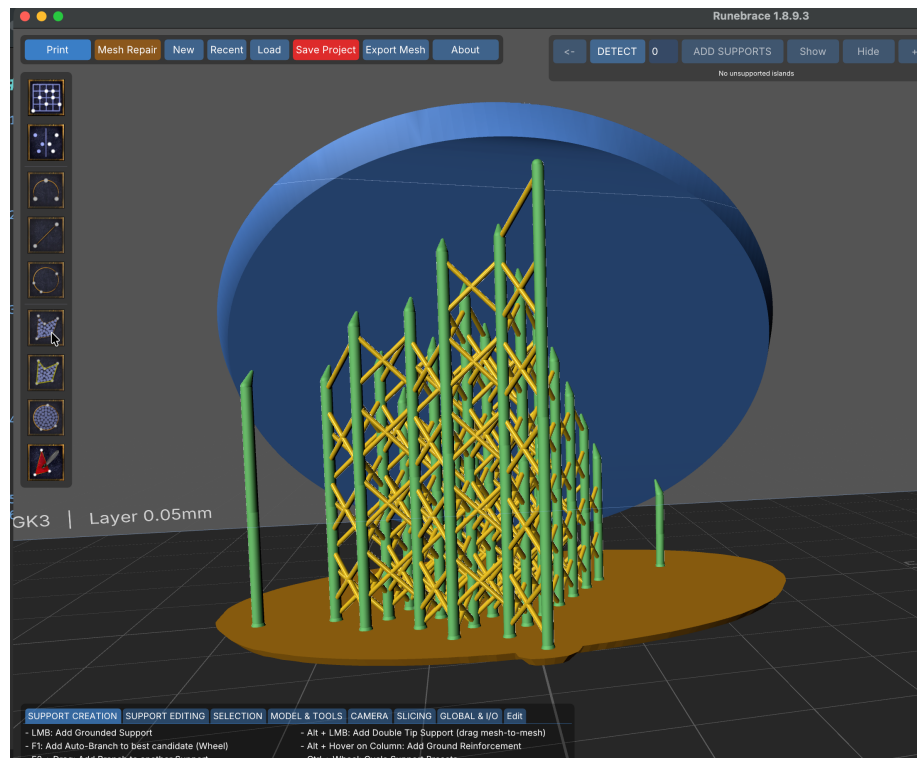
Fill an area defined by three or more supports.

Select three or more supports using F3 + LMB, then use the Poly Fill button on the left menu.



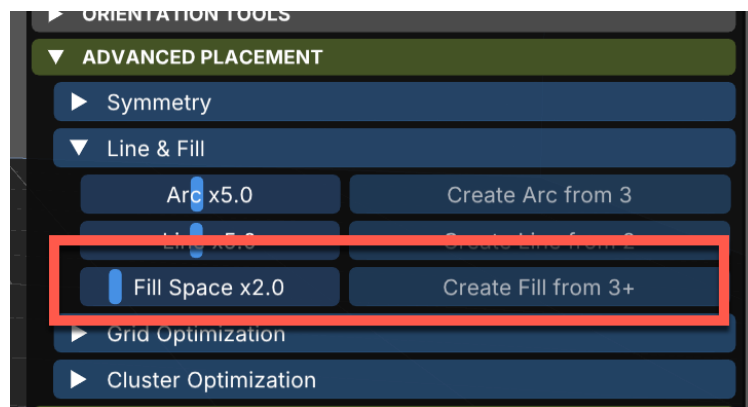
You can also use the Fill Space button from the right menu (see below).

Using Poly Fill/Fill Space from our starting point results in this:



Why: EDIT: ADD USE CASE

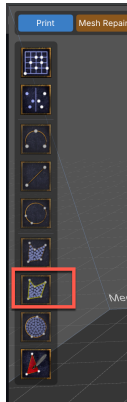
Adjust the spacing of the filled supports under Right Menu => Advanced Placement => Line and Fill => Fill Space



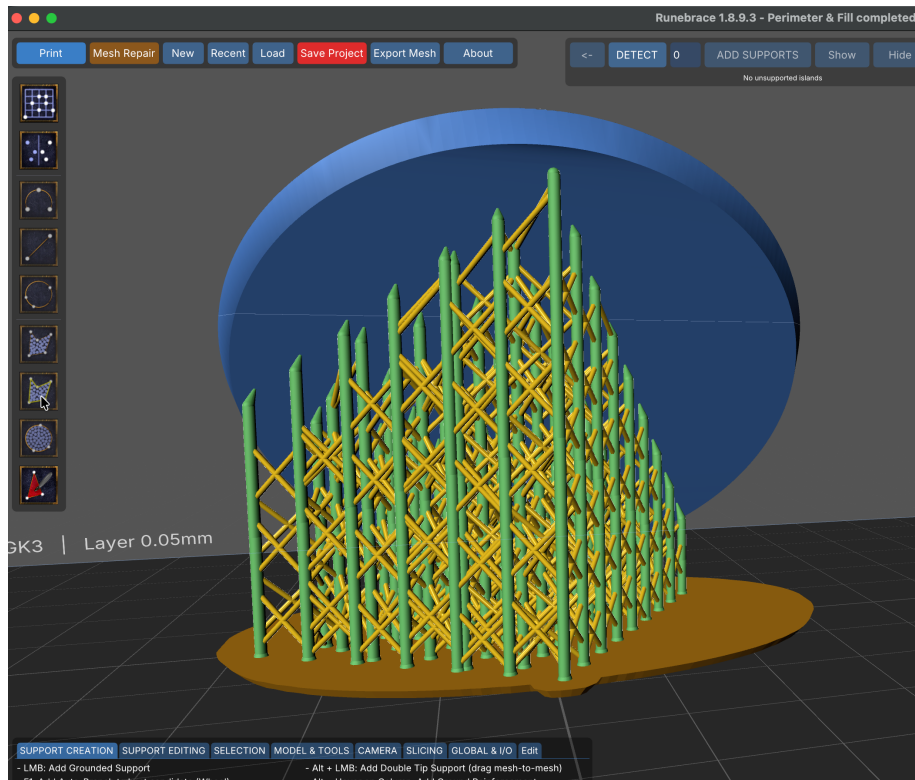
Perimeter and Fill

Perimeter & Fill does the same but also lays supports along the outline itself. Note the menu button's icon shows this via the yellow outline.

Select three or more supports, then press the Perimeter and Fill button (there's no alternative from the right menu). One action creates the perimeter and inside area supports.



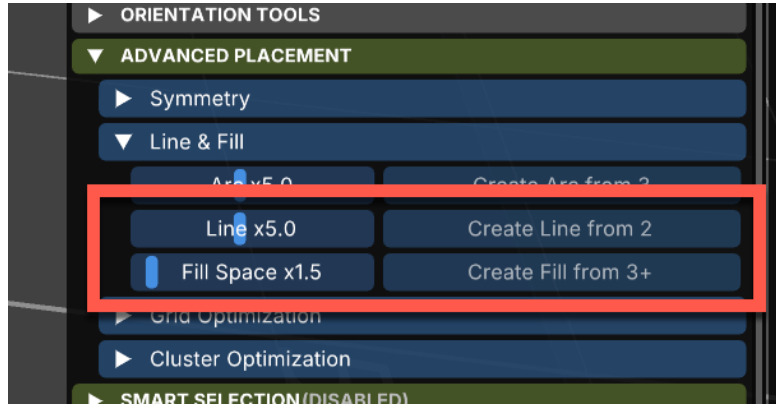
Using Perimeter and Fill from our starting point results in this:



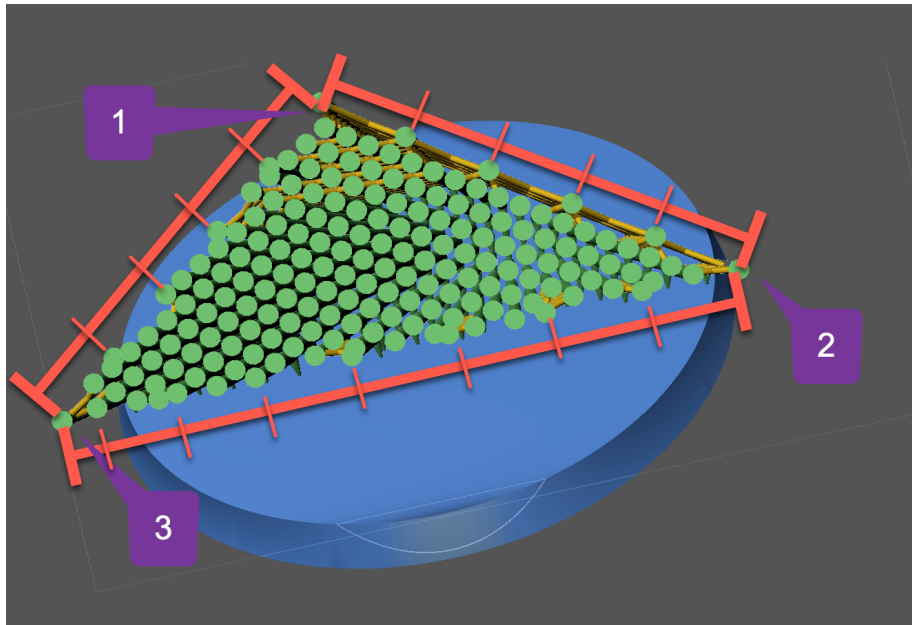
Perimeter and Fill has two separate adjustments:

- **Line:** Spacing between supports on the lines of the outer perimeter

- **Fill Space:** Spacing between supports in the grid that fills the area

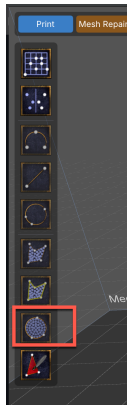


These settings are demonstrated in the example below. The perimeter supports are denoted by the red hash lines and were configured with a spacing of 5.0. The inside area was configured with a spacing of 1.5.

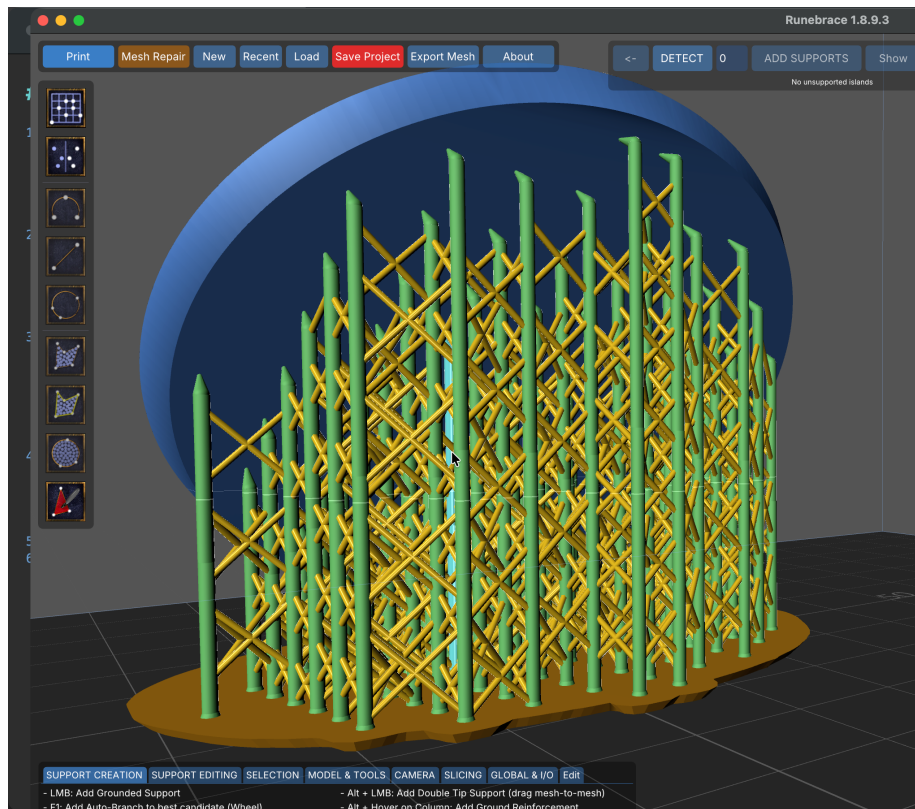


Circle Fill

Circle Fill takes exactly three supports, builds the circle that passes through their contact points, and fills that.



Using Circle Fill from our starting point results in this:



Triangle Painting

Rim

Deeper Dive on Placement

” Five details worth having in the docs. The perimeter walks the supports in selection order and closes the loop back to the first one, so the last segment is support N to support 1. The fill plane is defined by the first three supports only, not by all of them: any support beyond the third is flattened onto that plane, so a selection that is not roughly planar will not fill the way it looks. The grid covers the bounding box of the polygon in that plane and keeps the points that fall inside it, with one axis along support 1 to support 2 and the other perpendicular to it. Each grid point is then projected onto the model along the average tip direction of the selected supports, so the fill follows the surface, and where that ray misses the mesh no support appears, which is why gaps show up over holes or steep overhangs. Finally, Line and Fill Space are multipliers, not distances, and they multiply different reference diameters: Line multiplies the Base diameter, Fill Space multiplies the Top diameter. The same number on both sliders does not give the same spacing on the outline and on the fill. One more practical note: the operation records two undo steps, the perimeter and the fill, so undoing it takes two presses.”

“very small values stop having an effect below a certain point, and a segment too short to fit anything simply gets no supports in between. Happy to give you the exact arithmetic if you want a footnote.”

Math on Determining Placement “ $\text{step} = \text{Base diameter} \times \text{Line}$ (or $\text{Top diameter} \times \text{Fill Space}$). $\text{Gaps} = \text{segment length} \div \text{step}$, rounded down. $\text{Supports added} = \text{gaps} - 1$, spread evenly, so the real spacing is $\text{segment length} \div \text{gaps}$. Because the count rounds down, the spacing is always at least the step and can reach almost double it. Yours: $26.73 \text{ mm} \div 4 \text{ gaps} = 6.68 \text{ mm}$, the 6.64 you measured.”

More Complex Examples

You should now understand the basics of creating supports with the various tools. The example above is very simplistic, so let's use more complex

Presets

Working with Meshes

Hollowing

Printing and Slicing

Glossary

Here are common terms and phrases used in Runebrace.

Path Path applies to Line, Arc, and Circle. For two selected supports, Runebrace builds a path using a raycast between two points. For three selected supports, Runebrace will build a path by creating a curve that passes through all three supports' contact points.

Rim In triangle mode, “Rim” denotes the border of a painted triangle selection. It is determined from the mesh edges, and can have both an outer border *plus* the borders of any holes inside that particular selection. See also: “Outer Rim Only” and “All boundaries” options in the Triangle Paint panel.